Repurposing Learn-to-Sample for Multiclass Classification

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Abstract

Not Complete Yet

CCS Concepts

• **Computing methodologies** → **Supervised learning by classification**; *Neural networks*; • **Information systems** → Text classification.

Keywords

Learn-to-Sample, active learning, pseudo-labeling, large language models, multiclass classification, emotion classification, class imbalance

1 Introduction

Learn-to-Sample (LTS) is a cost-efficient, LLM-driven approach for building classifiers in settings where class imbalance makes it difficult to obtain relevant and labeled training data. In prior work [1], LTS has been used to identify wildlife trafficking in online marketplaces, demonstrating strong performance in generating pseudo-labels from small samples and training specialized binary classifiers across multiple case studies. However, LTS' effectiveness for multi-class classification problems remains unexplored. This paper addresses this gap by extending LTS to a multiclass problem and applying it to predict emotions from textual data with complex and imbalanced label structures.

Emotions Dataset [2]. The emotions dataset for NLP was collected from Twitter and pre-prepared for an investigation into emotion recognition [3], published on Kaggle. It contains textual data with corresponding emotions as labels. The label can be one of six emotions (the paper has 8): anger, fear, joy, love, sadness, or surprise. Said emotions occupy ~13, ~12, ~33, ~8, ~29, and ~3 percent of the training data respectively, making the dataset well-suited for examining LTS performance for multi-class classification and large class imbalances.

2 Related Work

The previously mentioned LTS exemplar involving wildlife trafficking identification outlines five steps in their LTS pipeline: create clusters, select cluster, select sample, LLM pseudo-labeling, and retrain model. Here, we will detail their approach and describe the conceptual framework of LTS which we have faithfully adopted with minor adaptations for our methodology discussed later on.

Create Clusters. LTS assigns each training data point to a cluster to ensure a diverse sample. The type of clustering and the corresponding number of clusters are chosen deliberately. For the exemplar, a Linear Discriminant Analysis (LDA) algorithm was used to cluster the training data; this allows the model to group similar ad titles in a way that captures structure in the feature space for more targeted sampling. The number of clusters (10) was chosen as a result of grid search.

Select Cluster. The exemplar's Thompson sampler uses a multi-armed bandit (MAB) strategy in which model performance following each training iteration informs whether to keep sampling from previously selected clusters. Previously chosen clusters are assigned a reward based on their contribution to model performance, modeled as Beta distributions whose parameters are updated to guide subsequent cluster selection. Clusters with stronger gains accrue higher likelihood to be selected in future iterations, while underperforming clusters are sampled less but are still explored due to posterior stochasticity.

Select Sample. A sample size of 200 is drawn from the selected cluster, initially randomly prior to a fine-tuned model, and subsequently through labels obtained from inference on the selected cluster by a fine-tuned model. Obtaining a sufficient number of positively labeled data - indicated by a parameter - is prioritized to ensure balance in the training data.

LLM Pseudo-Labeling. Once samples are obtained from the cluster, an LLM-based labeler assigns pseudo-labels to each sample using few-shot in-context learning. The labeler passes a prompt, a task description, and representative examples for each class to the LLM, which then returns a predicted class label for each sample serving as the training signal for the downstream classifier. These pseudo-labels have accuracies directly dependent on the quality of the prompt and the LLM used.

Retrain Model. The pseudo-labeled data is used to train a base model (bert-uncased-base) whose results are evaluated with respect to human-labeled validation data. If the chosen metric (weighted F1) improves relative to its performance (0.5 baseline), the Thompson sampler increments its win record for the relevant cluster. Conversely, if performance worsens after training, the Thompson sampler increments its loss record for the selected cluster. These five steps were repeated iteratively 10 times to produce a fine-tuned model with the highest score for the chosen metric (weighted F1).

3 Methodology

Data preparation. The emotions dataset from [2] contains three text files (train, test, validation). We converted all three into CSV files, the first two spliced together as training data and the latter as validation data for the BERT trainer. The classes 'anger', 'fear', 'joy', 'love', 'sadness', and 'surprise' were labeled alphabetically as 0 through 5.

Create Clusters. Whereas the exemplar used LDA for topical grouping, we opted to apply KMeans on normalized semantic embeddings generated by a pretrained SentenceTransformer model. This approach groups samples by contextual similarity in embedding space rather than relying on word co-occurrence patterns alone like in LDA, the latter of which was successful for its purpose but fails to capture textual connotation, an important detail for our classification problem. Shown in Figure 1, a Silhouette Score analysis based on clustered data indicated that a cluster size of 5 provided the best balance between cluster cohesion and separation.

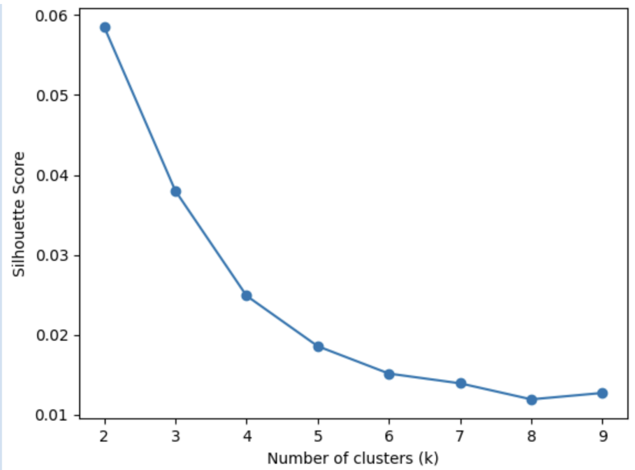

Figure 1: Silhouette Scores of KMeans Clusters on Normalized Semantic Embeddings from a Pretrained Model (BAAI/bge-large-en-v1.5).

Select Cluster. We propose a Dirichlet-Multinomial bandit framework for cluster selection, replacing the previously-used Thompson sampler with a multiclass alternative that assigns a Dirichlet concentration vector per cluster updated by per-class F1 deltas rather than collapsing per-class rewards as integer scalars for wins and losses. Following each model training iteration, the next bandit selection is guided by a weakness-weighted scoring function that aligns cluster selection with the model's current per-class deficiencies: this ensures that the labeling budget is allocated to clusters most likely to produce improvements where they are needed most. A temporal decay mechanism similar to that deployed in the Thompson sampler handles non-stationarity arising from model evolution across training iterations.

LLM Pseudo-Labeling. In the same manner as the exemplar, few-shot prompting was used to label textual data as one of six emotions. As opposed to making API calls to different LLM providers for different models, our approach relies on making requests to Ollama's Localhost server, requiring Ollama installation and prior to running our model, pulling different LLMs from Ollama's online repository onto a local device [4]. In addition, we parallelized the API call process through Dask, allowing for a shorter labeling time relative to the exemplar's original process. Furthermore, through trial and error, it was found that LLMs in some cases may not confidently label a text for any of the emotions provided: we implemented a failsafe feature to retry prompts and to pick a label randomly after 10 tries.

Retrain Model. We retain bert-base-uncased as the base model. While the exemplar uses a weighted F1 score to threshold updates to the Thompson sampler's wins and losses, our approach returns per-class F1 metrics, which are used to update class-specific labeling strength for the selected cluster in subsequent bandit selections. We then compute macro F1 to evaluate overall performance relative to prior iterations and use it as the criterion for updating the Dirichlet–Multinomial sampler.

4 Results

To evaluate whether LTS can be adapted from binary wildlife-trafficking detection to multiclass emotion classification, we tested our modified LTS pipeline on the Emotions dataset. Our results focus on four areas: dataset caveats, LLM baseline performance, LTS-text class-level performance, and cluster analysis.

4.1 Dataset Caveats

Before interpreting model performance, we inspected the Emotions dataset qualitatively and found several issues that make this task harder than a clean benchmark classification problem. These caveats are important because LTS relies on LLM-generated pseudo-labels, and certain types of text can make both the LLM labeler and the BERT classifier less reliable. This is especially relevant because our implementation includes a fallback for invalid LLM outputs: if the LLM fails to return one of the six allowed emotion labels after more than 10 attempts, the labeler randomly selects one said valid labels. In particular, examples involving sexual content or self-harm content may cause the LLM to refuse, avoid answering, or return something outside the required label set. As a result, these examples can introduce random pseudo-label noise into the training set when the fallback is triggered.

First, some examples contain sexual or romantic content. Examples are shown in Table 1. Some examples are more about sexual desire, embarrassment, inhibition, or discomfort. The main issue for LLM pseudo-labeling is that LLMs may sometimes refuse to respond to prompts involving sexual content or return an invalid response instead of one of the required emotion labels.

Table 1: Examples of sexual or romantic ambiguity in the Emotions dataset.

Example text	Label
"i am feeling horny so i ask her that lets go home"	love
"i know about have to do largely with the fact that any feelings romantic or sexual i have successfully hidden from myself"	love
"i feel inhibited by not having an outlet to deal with my sexual tensions"	sadness

Second, the dataset includes examples related to self-harm, suicidal feelings, or severe distress. Some examples are shown in Table 2. These examples are emotionally complex, but the bigger practical issue for the pipeline is that LLMs may avoid responding directly, refuse the prompt, or return an output that does not match the required label set. This means the pseudo-labeling step may fail even when the underlying emotion could be identified.

Table 2: Examples of severe distress or self-harm-related content.

Example text	Label
"ive learned how to turn off all my emotions more and more and i often find myself feeling completely blank while my mother is crying continuously over my suicidalness"	sadness
"i feel like i ve fucked up massively for not being able to fight off being suicidal"	anger
"i feel like hopeless helpless worthless scum"	sadness

In addition to sensitive content, some examples are emotionally ambiguous because they contain cues for more than one emotion. For example, the validation set includes the text "i am just feeling cranky and blue," which is labeled as anger. The word "cranky" suggests anger, but "blue" can suggest sadness. Since the dataset requires a single label, examples like this force the model to learn one emotion even when the text may reasonably express multiple emotions.

Finally, some entries appear unfinished, vague, or corrupted by preprocessing artifacts. Examples are shown in Table 3. These examples are problematic because they do not always contain enough emotional information for a human or model to confidently classify them. The last two examples also show leftover HTML or web artifacts such as `target blank`, `href`, `img`, and `nofollow`. These artifacts likely add noise to the text embeddings and may distract both the LLM labeler and the BERT classifier.

Table 3: Examples of vague, unfinished, or artifact-heavy text.

Example text	Label
"no description"	anger
"no response"	anger
"i feeling i should do fill in the blank"	sadness
"i feel horrible about wanting sonipro amp source geekparty linkedin a target blank title share on tumblr rel nofollow href http www"	sadness
"i feel studying and doing homework again after weeks of holidays target blank img title stumbleupon class ssba alt stumbleupon src http www"	sadness

Overall, these caveats show that the Emotions dataset is not only imbalanced, but also difficult for LLM-based pseudo-labeling because some examples contain content that may lead to invalid or missing LLM responses. This makes it a meaningful stress test for LTS. Unlike the original wildlife-trafficking task, where labels often depend on concrete product features, emotion classification requires the model to infer subjective emotional states from short and sometimes messy text while also handling cases where the LLM labeler may not return a usable answer.

4.2 LLM Baseline Performance

This result is more mixed than the original LTS paper's results. In the wildlife-trafficking setting, LTS-text often performed better than open-source LLMs and comparably to GPT-4. One reason for this difference is that the original LTS paper used GPT-4 for pseudo-labeling, so its trained classifier closely reflects GPT-4-level label quality. In this study, `glm4:9b` was specifically used for pseudo-labeling when running LTS; in turn, the resulting LTS-text classifier performed closer to that of `glm4:9b`. This reinforces the main idea behind LTS: even with minimal human-labeled data and an otherwise unlabeled training set, LLM pseudo-labeling can be used to train a reusable classifier. Once the BERT classifier is trained, future examples can be classified with a lightweight discriminative model rather than repeatedly running generative LLM inference.

Table 4: Performance comparison between direct LLM classifiers and the final LTS-text model.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
`llama3:8b`	0.4260	0.4343	0.4617	0.4012
`granite4.1:8b`	0.5760	0.5255	0.5220	0.4914
`glm4:9b`	**0.6240**	**0.5912**	0.5890	**0.5797**
LTS-text	0.5990	0.5681	**0.5916**	0.5661

The results show that LTS-text outperformed `llama3:8b` and `granite4.1:8b` in macro F1, but it performed slightly below `glm4:9b`. The best direct LLM baseline was `glm4:9b`, with a macro F1 score of 0.5797, while LTS-text reached 0.5661. This means LTS-text did not fully outperform every LLM baseline, but it came close to the strongest one while still using a smaller specialized classifier. This is not necessarily a failure of LTS; rather, it suggests that when the same local LLM used for pseudo-labeling is also a strong direct classifier, the downstream BERT model may preserve much of the LLM signal but not fully exceed it, especially when pseudo-labels are noisy.

4.3 LTS-Text Class-Level Performance

The final LTS-text model achieved an overall accuracy of 0.5990 and a macro F1 score of 0.5661 on the validation set. The class-level F1 scores are shown in Table 5.

Table 5: Class-level F1 scores for the final LTS-text model.

Class	Emotion	F1 Score	Training Share (%)
0	Anger	0.5619	13.5
1	Fear	0.5267	12.1
2	Joy	**0.7320**	33.5
3	Love	0.4241	8.2
4	Sadness	0.5442	29.2
5	Surprise	0.6076	3.6

The strongest class was joy, with an F1 score of 0.7320. This is not surprising because joy is the largest class in the training data, making up about 33.5% of the examples. The weakest class was love, with an F1 score of 0.4241. Love is one of the least represented classes, making up about 8.2% of the training set. It is also somewhat close to joy and sometimes appears in romantic, relationship-related, or sexual examples that may cause LLM responses to become less reliable, which likely makes it harder to classify consistently.

Surprise is interesting because it is the smallest class, making up only about 3.6% of the training set, but it still reached an F1 score of 0.6076. This suggests that class frequency alone does not explain performance. It also suggests that our multiclass sampler may be helping: unlike the original LTS setup, which used weighted F1, our sampler prioritizes classes with weaker per-class F1 scores when choosing future clusters to sample from. The relatively strong performance on surprise, despite its low frequency, indicates that this weakness-guided sampling strategy may help the model recover performance on underrepresented classes. Surprise may also have more distinctive language patterns than love.

4.4 Cluster Analysis

Our modified LTS pipeline used KMeans clustering over normalized semantic embeddings instead of the original paper's LDA-based clustering. This choice was motivated by the fact that emotion classification depends on contextual similarity and tone, not only topic-level word co-occurrence. A silhouette score analysis supported using five clusters for the KMeans setup.

The cluster distribution also suggests that the clusters captured meaningful differences in the data. The class-cluster distribution is shown in Table 6.

Table 6: Training data class distribution across five KMeans clusters.

Emotion	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Anger	98	307	1078	131	545
Fear	99	1214	154	98	372
Joy	2096	667	282	1905	412
Love	171	178	158	716	81
Sadness	236	676	1660	268	1826
Surprise	85	211	39	140	97

This table shows that labels are not evenly spread across clusters. For example, fear appears heavily in Cluster 1, sadness appears heavily in Clusters 2 and 4, and love appears most strongly in Cluster 3. This supports the idea that cluster-guided sampling can help LTS find useful regions of the dataset instead of sampling randomly.

However, the table also shows why multiclass LTS is harder than binary LTS. In the original binary setting, the sampler mostly needed to find enough positive examples while still keeping the training set diverse. For our purpose, the sampler must balance six classes at once. Sampling from a cluster that helps one weak class may not help another. This motivated the use of a multiclass bandit-style sampler that updates sampling behavior based on per-class performance rather than a single binary win/loss score.

4.5 Discussion

Overall, the results suggest that LTS can be adapted to multiclass emotion classification, but the benefits are less straightforward than in the original binary wildlife-trafficking setting. LTS-text achieved competitive performance and beat two of the three direct LLM baselines, but it did not surpass `glm4:9b`, the model used for pseudo-labels. This suggests that LTS is promising, but direct LLM classification remains a strong baseline for subjective tasks like emotion detection.

The main difficulty is that emotion labels are not as concrete as wildlife product labels. In the original paper, a product either contains an animal-derived material or it does not, even if the advertisement is noisy. In the Emotions dataset, a text can express multiple emotions at once, and the correct label may depend on subtle context. This creates problems for both pseudo-labeling and model training.

The dataset caveats also help explain the final performance. Sexual content and self-harm-related content can cause LLMs to refuse, avoid answering, or return invalid labels, which may trigger randomness through the labeling pipeline fallback. Unfinished or corrupted examples can also provide too little useful signal. These issues make the task especially difficult for a single-label classifier trained on pseudo-labels.

Even with these limitations, the LTS-text model's performance is meaningful. The model reached a macro F1 close to the best local LLM baseline while remaining reusable and cheaper to run after training. This aligns with the broader motivation of LTS: use LLMs selectively for pseudo-labeling, then train a smaller classifier that can be applied repeatedly without querying an LLM for every new example.

5 Conclusion

This paper explores whether Learn-to-Sample, originally designed for cost-effective binary classification in wildlife-trafficking detection, can be repurposed for multiclass emotion classification. We adapted the original LTS pipeline by using clustering over semantic embeddings and by introducing a multiclass sampling strategy based on class-specific performance. These changes were necessary because emotion classification involves multiple classes rather than a single positive/negative distinction.

Our results show that LTS can be extended to a multiclass setting in implementation, but its performance gains are more limited than in the original binary wildlife-trafficking task. The final LTS-text model achieved 0.5990 accuracy and 0.5661 macro F1. It outperformed `llama3:8b` and `granite4.1:8b`, but fell slightly below `glm4:9b`. This means that LTS did not clearly dominate all direct LLM baselines, but it still produced a competitive reusable classifier.

The class-level results showed that joy was the easiest emotion to classify, while love was the hardest. This likely reflects both class imbalance and semantic overlap between emotion categories. Love was relatively rare and often overlapped with joy, sadness, and sexual or romantic language. Surprise, despite being the rarest class, performed better than love, suggesting that separability of language patterns matters in addition to class frequency.

A major limitation is that the final classifier is bounded by the quality of the pseudo-labels used during training. Invalid LLM outputs, fallback random labels, and missing responses caused by sensitive content can all weaken the training signal. The dataset itself introduced important limitations: some examples contained sexual content, self-harm-related content, vague text, or web artifacts left over from preprocessing. These examples can make LLM pseudo-labeling less reliable because the model may refuse to answer, return invalid labels, or receive too little useful signal from the text. This suggests that future work should include stronger data cleaning, error analysis, and possibly a multi-label setup where one text can express more than one label.

Overall, our findings suggest that LTS is adaptable beyond binary wildlife-trafficking detection, but multiclass emotion classification exposes new challenges. The approach is most useful when LLM pseudo-labels are reliable and when the target classes are separable. For subjective emotion labels, LTS still produces a competitive reusable classifier, but direct LLM classification remains a strong baseline and pseudo-label noise becomes a more important bottleneck. Future work could test stronger LLM labelers, compare against random and stratified sampling baselines, add more robust preprocessing, and evaluate whether multi-label emotion classification improves performance.

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